

Learning Objectives

By the end of this lecture, the student will be able to

1. Describe the difference between different types of hypersensitivity reactions
2. Describe some of the physical exam findings in immediate hypersensitivity reactions
3. Describe some of the physical exam findings in atopy
4. Select appropriate laboratory diagnostic tests for evaluation of patients with hypersensitivity reactions
5. Select or identify appropriate treatments for immediate hypersensitivity reactions

Case Study: Jack

A 4-year-old boy named Jack has been brought into the ED with swelling of the face, generalized itchiness, and drowsiness. The symptoms developed a few minutes after eating a chocolate bar. His mother explains that he has had minor reactions to pistachios in the past, including lip swelling. The boy also has eczema, and his mother thinks that in the past, this has flared up when he has eaten hazelnuts. She has concluded that he might be allergic to tree nuts and has tried to eliminate them from the house. The patient has never had any symptoms from milk. Jack asks if you're going to take away the itch, and if he can have more chocolate when he gets home.

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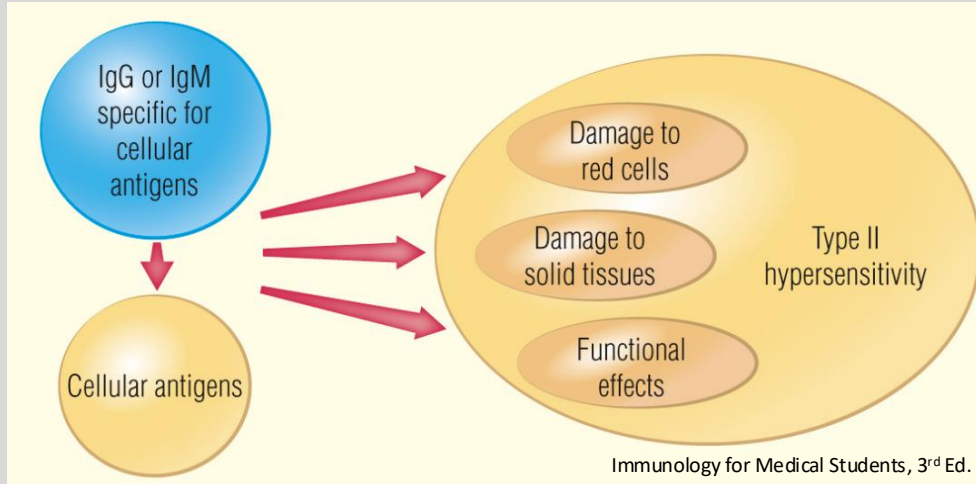


Immunology for Medical Students, 3rd Ed.



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Type II Hypersensitivity

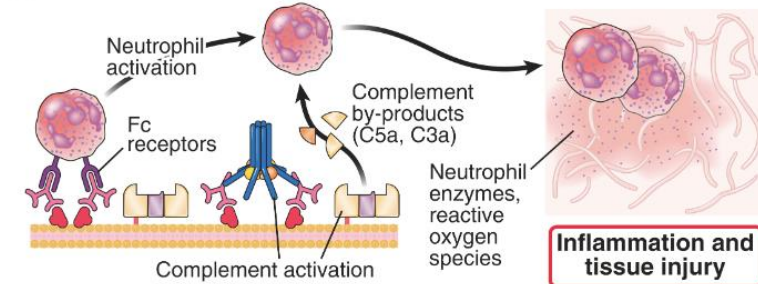


Cellular destruction: autoimmune hemolytic anemia,

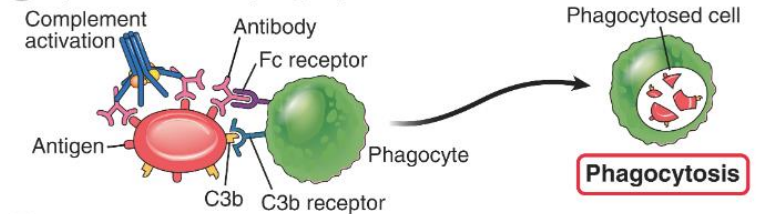
Inflammation: rheumatic fever, hyperacute transplant rejection, Goodpasture syndrome

Cellular dysfunction: Myasthenia gravis, Graves disease

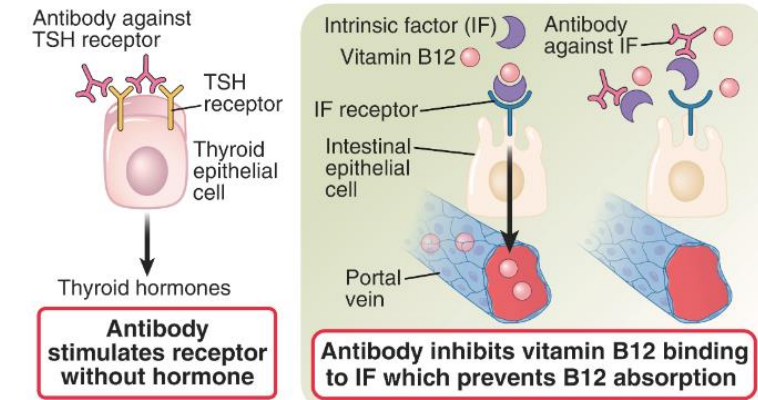
A Complement- and Fc receptor-mediated inflammation



B Opsonization and phagocytosis



C Abnormal physiologic responses without cell/tissue injury



Type III Hypersensitivity

Serum sickness: **systemic** inflammatory disease with fever, rash, arthritis

- Production of antibodies against foreign antigen → 1-2 weeks later immune complexes form and deposit in tissues

Arthus reaction: **local** vasculitis

- Intradermal injection of antigen into pre-sensitized individual leads to immune complex formation in skin

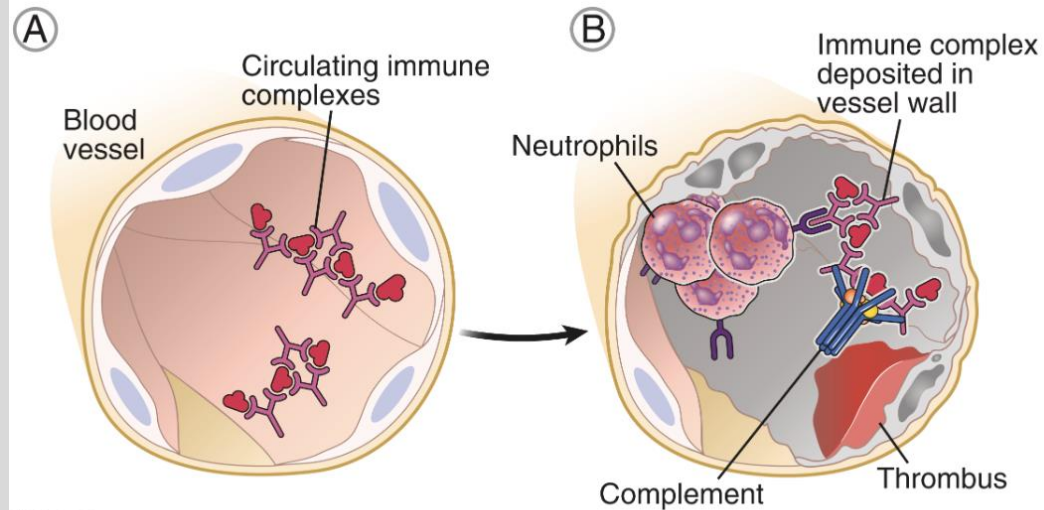
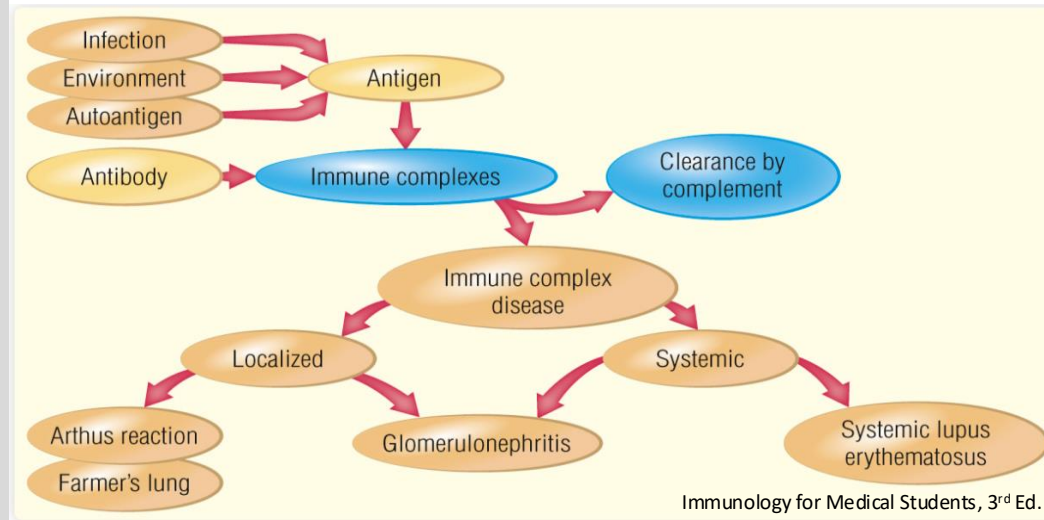


FIG. 11.9

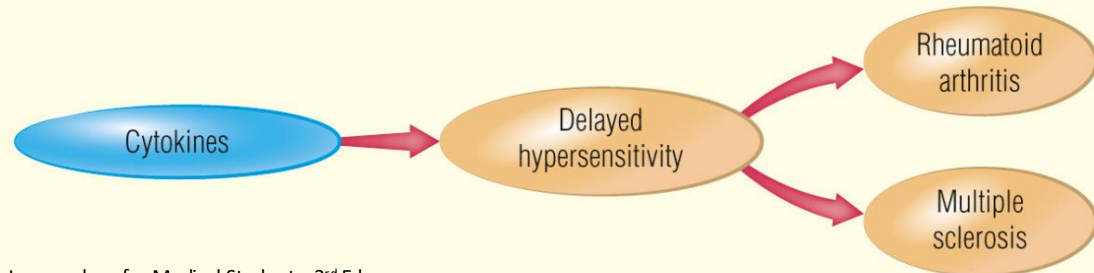
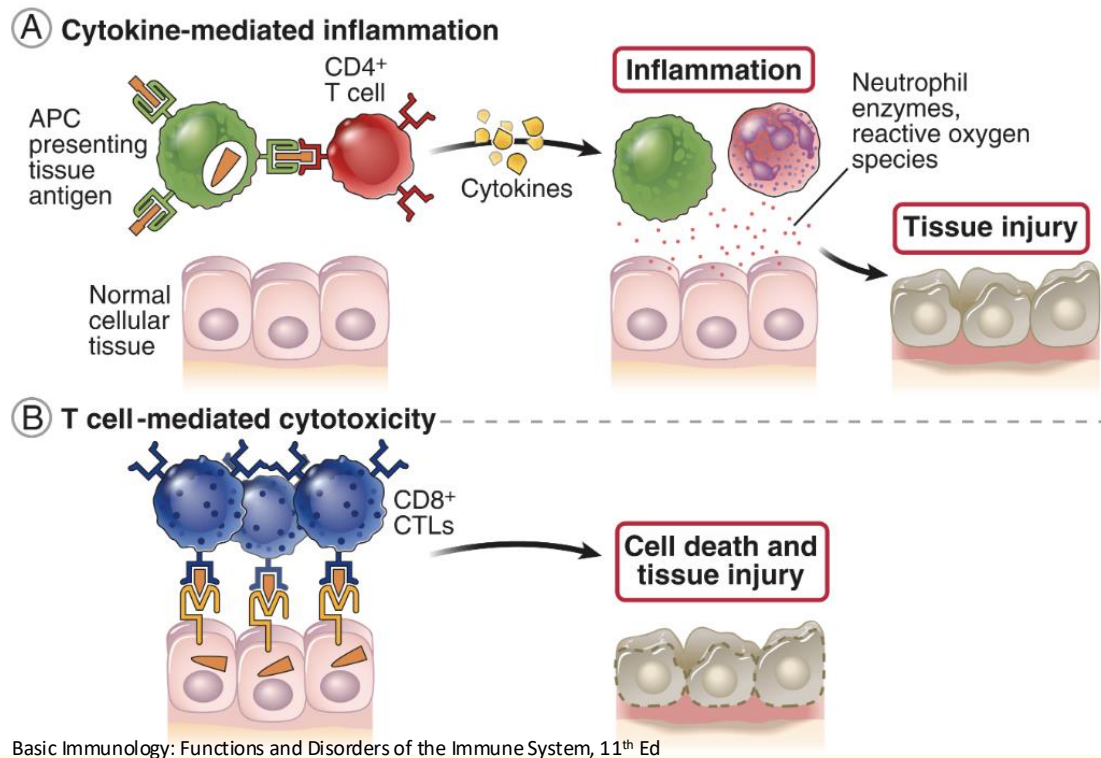
Basic Immunology: Functions and Disorders of the Immune System, 11th Ed



Type IV Hypersensitivity

Occurs **24 to 48**
hours after antigen
exposure/challenge

4T's: T cells, Transplant rejections,
Tuberculosis skin test, Touching
(contact dermatitis)



Immediate Hypersensitivity: Anaphylaxis

1. NO known allergen exposure + sudden onset illness with skin/mucosal involvement + either respiratory involvement or cardiovascular involvement
2. Likely or known allergen exposure + sudden onset of TWO or more of the following: skin/mucosal involvement, respiratory involvement, cardiovascular involvement, severe GI involvement
3. Known allergen exposure + sudden onset of EITHER respiratory involvement after exposure to a non-inhaled allergen OR cardiovascular involvement

Immediate Hypersensitivity: Anaphylaxis

Skin/Mucosal: urticaria, flushing, erythema, facial/lip/tongue/oropharyngeal swelling or itching, difficulty swallowing, mottling (infants), repetitive lip licking (infants)

Respiratory: wheezing, increased work of breathing, hypoxemia, cough, dyspnea, stridor, voice change, hoarse cry (infants)

Cardiovascular : hypotension, syncope, dizziness, altered mental status, persistent unexplained tachycardia (infants)

Severe GI: crampy abdominal pain, repetitive vomiting, diarrhea

Immediate Hypersensitivity: Anaphylaxis

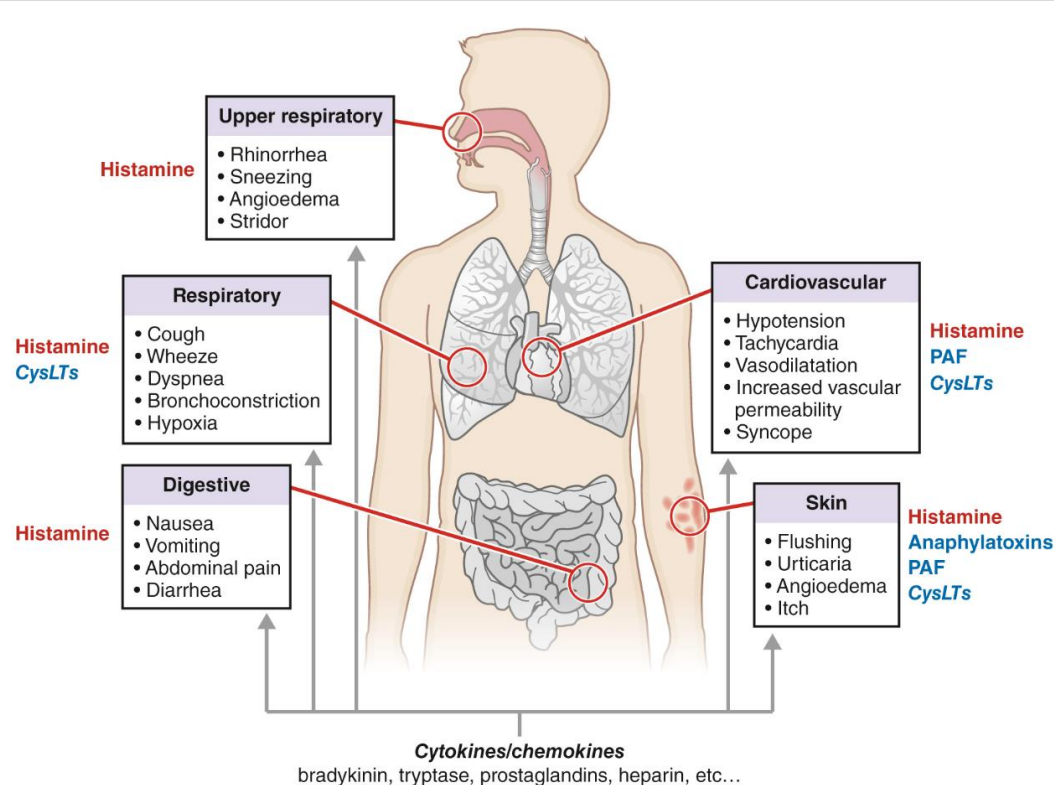


FIG. 75.3

Etiologies:

- Medications
- Foods
- Animal/Human Proteins
- Insect Venoms
- Vaccines
- Latex Products
- IV Contrast Media/Dyes/Colorants
- Enzymes
- Hormones
- Allergy Testing
- Allergen-Specific Immunotherapy
- Idiopathic

Anaphylaxis Management

1. Remove exposure to the allergen.
2. Assess ABCs (airway, breathing, circulation) + mentation AND
2. Administer epinephrine intramuscularly to anterolateral thigh (vastus lateralis m.) or intranasally.
3. CALL 9-1-1. **ALL PERSONS WHO HAVE BEEN ADMINISTERED EPINEPHRINE MUST GO TO THE ED.**
4. If ABCs are acceptable, place the patient in the supine position and elevate lower extremities.
5. CONTINUOUSLY EVALUATE ABCs.



Anaphylaxis Management

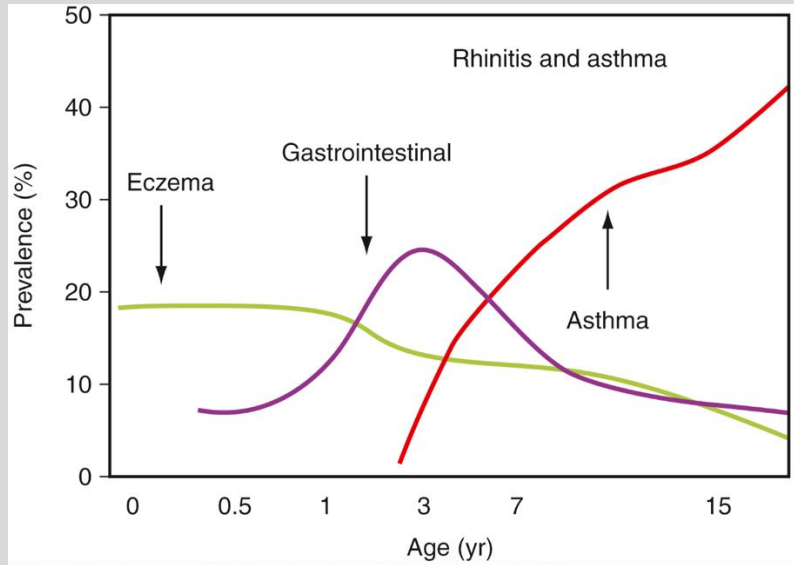
When to Observe for Prolonged Period (8-24h)/Admit:

1. Moderate to severe reaction (hypoxemia, airway edema, shock)
2. Required more than 1 dose of epinephrine
3. Prolonged duration of symptoms (>4 hours)
4. History of severe asthma or wheezing
5. History of severe or biphasic reactions
6. Ingested antigen with possibility of continued absorption

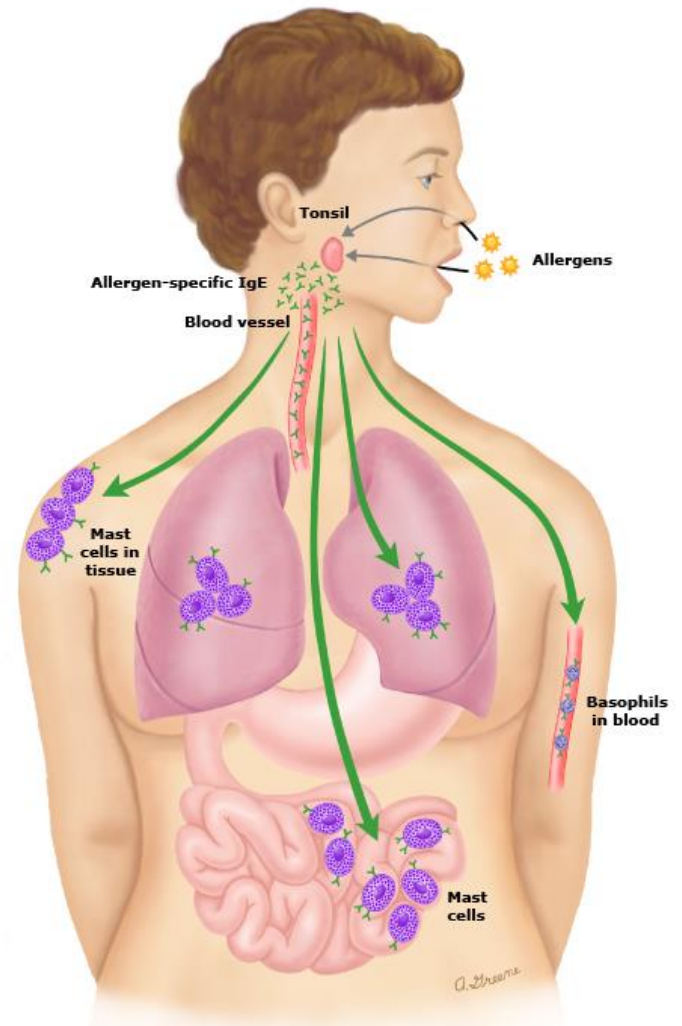
What is Atopy?

“Allergic triad”/“Allergic march”/“Atopic march”

Atopic dermatitis in infancy **PROGRESSES TO** childhood/young adult asthma + allergic rhinitis



Henry's Clinical Diagnosis and Management by Laboratory Methods, 24th Ed.



The relationship between IgE and allergic disease, UpToDate

Asthma

National Asthma Education and Prevention Program: common chronic disorder of the airways that is complex and characterized by variable and recurring symptoms, **airflow obstruction**, bronchial hyperresponsiveness, and an underlying **inflammation**.

Global Initiative for Asthma: a heterogeneous disease, usually characterized by chronic airway **inflammation**. It is defined by the history of respiratory symptoms such as wheeze, shortness of breath, chest tightness, and cough that vary over time and in intensity, together with variable **expiratory airflow limitation**.

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Summary

- Type I/Anaphylaxis/Atopic Hypersensitivity reactions are caused by degranulation of mast cells and eosinophils
- Type II/Antibody-mediated Hypersensitivity reactions are caused by IgG reacting with complement or Fc receptor on macrophages
- Type III/Immune complex Hypersensitivity reactions are caused by immune complexes of antigen and antibody that cause damages at the site of production (Arthus) or circulate and cause damage elsewhere (serum sickness)
- Type IV/Delayed Hypersensitivity reactions are caused directly by CD8+ cytotoxic T-cells or indirectly by effector CD4+ T-cells which initiate release of inflammation-inducing cytokines

Summary (cont'd)

- Anaphylaxis is a serious hypersensitivity reaction that can progress rapidly and may cause death.
- Epinephrine is the first line treatment for anaphylaxis and should be administered as early as possible if anaphylaxis is suspected.
- Atopy is a genetic propensity to produce allergen-specific IgE.
- Allergy is diagnosed when individuals have both allergen-specific IgE and symptoms upon exposure to that allergen.
- Atopic march/allergic march/allergic triad: atopic dermatitis + allergic rhinitis + asthma
- Diagnosis of hypersensitivity can consist of intradermal testing, patch/prick testing, blood testing, and/or pulmonary function testing.

Question #1

A 34-year-old woman is brought to the emergency department via EMS after collapsing at a work picnic in the park. She reports diffuse, crampy abdominal pain with one episode of vomiting on the way to the hospital. Temperature is 99.1F, pulse is 121 bpm, and blood pressure is 88/75mmHg. Oxygen saturation is 85% on room air. She is using accessory muscles to breathe and has high-pitched sounds on inspiration, and she has swelling around the eyes. Which of the following is most directly involved in causing this patient's symptoms?

- A. Bradykinin excess
- B. Histamine
- C. IgG autoantibodies
- D. Immune complexes
- E. Natural killer cells
- F. Th1 lymphocytes

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Common presentations of anaphylaxis include hypotension, hives, pruritic and/or flushed skin, swollen lips or tongue, periorbital edema, and conjunctival swelling. Anaphylaxis is a Type I hypersensitivity reaction characterized by binding of antigen to bound IgE on mast cells and basophils, followed by degranulation of mast cells and release of vasoactive amines like histamine and serotonin.

Question #2

A 65-year-old male comes to the family medicine clinic for evaluation of pruritic, erythematous papules, vesicles and blisters in a linear and geometric pattern on the lower extremities that appeared two days ago. He reports that he has been wearing shorts for the past week now that the weather is warmer outside. His daily activities include a 3-mile morning walk with his wife, reading, gardening, and playing chess with his friends. He denies use of new soaps or lotions, recent travel, and sick contacts. Which of the following mechanisms is most important in the pathogenesis of this patient's most likely condition?

- A. Activation of sensitized B lymphocytes
- B. Antibodies binding to cell surface antigens
- C. Antibody-antigen complex deposition
- D. Differentiation of CD4+ T lymphocytes into helper T lymphocytes
- E. Leukotriene-mediated inflammation



Question #2

Which of the following mechanisms is most important in the pathogenesis of this patient's most likely condition?

- A. Activation of sensitized B lymphocytes
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- E. Leukotriene-mediated inflammation

This patient is most likely experiencing an allergic contact dermatitis due to poison ivy, which he likely contacted while gardening. The allergic response to poison ivy is a type IV hypersensitivity reaction which requires secondary exposure to the antigen, which triggers differentiation of CD4+ T lymphocytes into T-helper type lymphocytes.